Juan G. Victores

Journal Articles (30)

1. Ana Calzada-Garcia, Juan G. Victores, Francisco J. Naranjo-Campos, and Carlos Balaguer. Inverse kinematics for robotic manipulators via deep neural networks: Experiments and results. *Applied Sciences*, 15, 6 2025a. ISSN 2076-3417. doi: 10.3390/app15137226. URL https://doi.org/10.3390/app15137226 [robot] [kinematics] (Q1)

2. Johnny J. Yepez-Figueroa, Juan G. Victores, Edwin Daniel Oña, Carlos Balaguer, and Alberto Jardón. Design and development of an omnidirectional three-wheeled industrial mobile robot platform. *Applied Sciences*, 15, 2025. ISSN 2076-3417. doi: 10.3390/app15105277. URL https://doi.org/10.3390/app15105277 [robot] [design] (Q1)

3. Ana Calzada-Garcia, Juan G. Victores, Francisco J. Naranjo-Campos, and Carlos Balaguer. A review on inverse kinematics, control and planning for robotic manipulators with and without obstacles via deep neural networks. *Algorithms*, 18, 1 2025b. ISSN 1999-4893. doi: 10.3390/a18010023. URL https://doi.org/10.3390/a18010023 [robot] [kinematics] (Q2)

4. Francisco J. Naranjo-Campos, Juan G. Victores, and Carlos Balaguer. Expert-trajectory-based features for apprenticeship learning via inverse reinforcement learning for robotic manipulation. *Applied Sciences*, 14, 11 2024a. ISSN 2076-3417. doi: 10.3390/app142311131. URL https://doi.org/10.3390/app142311131 [robot] [assistive] (Q1)

5. Francisco J. Naranjo-Campos, Juan G. Victores, and Carlos Balaguer. Method for bottle opening with a dual-arm robot. *Biomimetics*, 9, 9 2024b. ISSN 2313-7673. doi: 10.3390/biomimetics9090577. URL https://doi.org/10.3390/biomimetics9090577 [robot] [assistive] (Q2)

6. Francisco J. Naranjo-Campos, Ainhoa De Matías-Martínez, Juan G. Victores, José A. Gutiérrez Dueñas, Almudena Alcaide, and Carlos Balaguer. Assistance in picking up and delivering objects for individuals with reduced mobility using the tiago robot. *Applied Sciences*, 14, 8 2024c. ISSN 2076-3417. doi: 10.3390/app14177536. URL https://doi.org/10.3390/app14177536 [robot] [assistive] (Q1)

7. Stephen Fox and Juan G. Victores. Safety of human–artificial intelligence systems: Applying safety science to analyze loopholes in interactions between human organizations, artificial intelligence, and individual people. *Informatics*, 11:36, 2024. doi: 10.3390/informatics11020036. URL https://doi.org/10.3390/informatics11020036 [aml] (Q2)

8. Rubén de-la Torre, Edwin Daniel Oña, Juan G. Victores, and Alberto Jardón. Spasticsim: a synthetic data generation method for upper limb spasticity modelling in neurorehabilitation. *Scientific Reports*, 14:1646, 1 2024. ISSN 2045-2322. doi: 10.1038/s41598-024-51993-w. URL https://doi.org/10.1038/s41598-024-51993-w [robot] [rehabilitation] (Q2)

9. Raul Fernandez-Fernandez, Bartek Łukawski, Juan G. Victores, and Claudio Pacchierotti. Transferring human emotions to robot motions using neural policy style transfer. *Cognitive Systems Research*, 82:101121, 12 2023a. ISSN 1389-0417. doi: 10.1016/J.COGSYS.2023.05.010. URL https://doi.org/10.1016/J.COGSYS.2023.05.010 [robot] [xgnitive: cgda] (Q2)

10. Raul Fernandez-Fernandez, Juan G. Victores, and Carlos Balaguer. Deep robot sketching: An application of deep q-learning networks for human-like sketching. *Cognitive Systems Research*, 81:57–63, 9 2023b. ISSN 1389-0417. doi: 10.1016/J.COGSYS.2023.05.004. URL https://doi.org/10.1016/J.COGSYS.2023.05.004 [robot] [xgnitive: cgda] (Q2)

11. Raul Fernandez-Fernandez, Juan G. Victores, Jennifer J. Gago, David Estevez, and Carlos Balaguer. Neural policy style transfer. *Cognitive Systems Research*, 72:23–32, 3 2022a. ISSN 1389-0417. doi: 10.1016/j.cogsys.2021.11.003. URL https://doi.org/10.1016/j.cogsys.2021.11.003 [robot] [xgnitive: cgda] (Q2)

12. Andrea Gil Ruiz, Juan G. Victores, Bartek Łukawski, and Carlos Balaguer. Design of an active vision system for high-level isolation units through q-learning. *Applied Sciences*, 10:5927, 8 2020. ISSN 2076-3417. doi: 10.3390/app10175927. URL https://doi.org/10.3390/app10175927 [robot] [textiles: horus] (Q2)

13. David Estevez, Juan G. Victores, Raul Fernandez-Fernandez, and Carlos Balaguer. Enabling garment-agnostic laundry tasks for a robot household companion. *Robotics and Autonomous Systems*, 123: 103330, 1 2020. ISSN 0921-8890. doi: 10.1016/j.robot.2019.103330. URL https://doi.org/10.1016/j.robot.2019.103330 [robot] [textiles: folding] [textiles: ironing] (Q2)

14. Alice Stazio, Juan G. Victores, David Estevez, and Carlos Balaguer. A study on machine vision techniques for the inspection of health personnels' protective suits for the treatment of patients in extreme isolation. *Electronics*, 8:743, 6 2019. doi: 10.3390/electronics8070743. URL https://doi.org/10.3390/electronics8070743 [robot] [textiles: horus] (Q2)

15. Jennifer J. Gago, Juan G. Victores, and Carlos Balaguer. Sign language representation by teo humanoid robot: End-user interest, comprehension and satisfaction. *Electronics*, 8:57, 1 2019a. ISSN 2079-9292. doi: 10.3390/electronics8010057. URL https://doi.org/10.3390/electronics8010057 [robot] [sign-language] (Q2)

16. Raul Fernandez-Fernandez, Juan G. Victores, David Estevez, and Carlos Balaguer. Real evaluations tractability using continuous goal-directed actions in smart city applications. *Sensors*, 18:3818, 11 2018a. ISSN 1424-8220. doi: 10.3390/s18113818. URL https://doi.org/10.3390/s18113818 [robot] [xgnitive: cgda] (Q1)

17. Santiago Martinez, Juan Miguel Garcia-Haro, Juan G. Victores, Alberto Jardon, and Carlos Balaguer. Experimental robot model adjustments based on force-torque sensor information. *Sensors*, 18:836, 3 2018. ISSN 14248220. doi: 10.3390/s18030836. URL https://doi.org/10.3390/s18030836 [robot] [humanoid] (Q1)

18. Elisabeth Menendez, Juan G. Victores, Roberto Montero, Santiago Martínez, and Carlos Balaguer. Tunnel structural inspection and assessment using an autonomous robotic system. *Automation in Construction*, 87:117–126, 3 2018. ISSN 09265805. doi: 10.1016/j.autcon.2017.12.001. URL https://doi.org/10.1016/j.autcon.2017.12.001 [robot] [construction] (Q1)

19. Eugenio Marinetto, Juan G. Victores, Mónica García-Sevilla, Mercedes Muñoz, Felipe Ángel Calvo, Carlos Balaguer, Manuel Desco, and Javier Pascau. Technical note: Mobile accelerator guidance using an optical tracker during docking in ioert procedures. *Medical Physics*, 44:5061–5069, 2017. ISSN 2473-4209. doi: 10.1002/mp.12482. URL https://doi.org/10.1002/mp.12482 [robot] [medical] (Q1)

20. Konstantinos Loupos, Anastasios D Doulamis, Christos Stentoumis, Eftychios Protopapadakis, Konstantinos Makantasis, Nikolaos D Doulamis, Angelos Amditis, Philippe Chrobocinski, Juan G. Victores, Roberto Montero, Elisabeth Menendez, Carlos Balaguer, Rafa Lopez, Miquel Cantero, Roman Navarro, Alberto Roncaglia, Luca Belsito, Stephanos Camarinopoulos, Nikolaos Komodakis, and Praveer Singh. Autonomous robotic system for tunnel structural inspection and assessment. *International Journal of Intelligent Robotics and Applications*, pages 1–24, 2017. ISSN 2366-598X. doi: 10.1007/s41315-017-0031-9. URL https://doi.org/10.1007/s41315-017-0031-9 [robot] [construction]

21. David Estevez, Juan G. Victores, Santiago Morante, and Carlos Balaguer. Robot devastation: Using diy low-cost platforms for multiplayer interaction in an augmented reality game. *EAI Endorsed Transactions on Collaborative Computing*, 15:1–5, 2015a. doi: 10.4108/icst.intetain.2015.259753. URL https://doi.org/10.4108/icst.intetain.2015.259753 [robot] [video-game]

22. Santiago Morante, Juan G. Victores, and Carlos Balaguer. Cryptobotics: Why robots need cyber safety. *Frontiers in Robotics and AI*, 2:1–4, 2015a. doi: 10.3389/frobt.2015.00023. URL https://doi.org/10.3389/frobt.2015.00023 [robot] [cryptography]

23. Roberto Montero, Juan G. Victores, Santiago Martínez, Alberto Jardón, and Carlos Balaguer. Past, present and future of robotic tunnel inspection. *Automation in Construction*, 59:99–112, 2015a. doi: 10.1016/j.autcon.2015.02.003. URL https://doi.org/10.1016/j.autcon.2015.02.003 [robot] [construction] (Q1)

24. Santiago Morante, Juan G. Victores, Alberto Jardón, and Carlos Balaguer. Humanoid robot imitation through continuous goal-directed actions: An evolutionary approach. *Advanced Robotics*, 29:303–314, 2015b. ISSN 1568-5535. doi: 10.1080/01691864.2014.964314. URL https://doi.org/10.1080/01691864.2014.964314 [robot] [xgnitive: cgda] (Q4)

25. Juan G. Victores, Santiago Morante, Alberto Jardón, and Carlos Balaguer. An accessible interface for programming an assistive robot. *Journal of Accessibility and Design for All (JACCES)*, 4:161–176, 2014a. ISSN 2013-7087. doi: 10.17411/jacces.v4i3.49. URL https://doi.org/10.17411/jacces.v4i3.49 [robot] [assistive]

26. Jonathan Crespo, Ramon Barber, Juan G. Victores, and Alberto Jardón. Algorithm for graph visibility obtainment from a map of non-convex polygons. *Journal of Mechanical Engineering and Robotics Research*, 3:150–170, 2014. ISSN 2278-0149. URL https://www.ijmerr.com/v3n2/ijmerr_v3n2_19.pdf [robot] [planning]

27. Santiago Martínez, Alberto Jardón, Juan G. Victores, and Carlos Balaguer. Flexible field factory for construction industry. *Assembly Automation*, 33:175–183, 2013. doi: 10.1108/01445151311306708. URL https://doi.org/10.1108/01445151311306708 [robot] [construction] (Q4)

28. Alberto Jardón, Juan G. Victores, Santiago Martínez, and Carlos Balaguer. Experience acquisition simulator for operating microtuneling boring machines. *Automation in Construction*, 23:33–46, 2012a. doi: 10.1016/j.autcon.2011.12.002. URL https://doi.org/10.1016/j.autcon.2011.12.002 [robot] [construction] (Q1)

29. Alberto Jardón, Juan G. Victores, Santiago Martínez, Antonio Giménez, and Carlos Balaguer. Personal autonomy rehabilitation in home environments by a portable assistive robot. *IEEE Trans. on Systems, Man, and Cybernetics, Part C: Applications and Reviews*, 42:561–570, 2011a. doi: 10.1109/TSMCC.2011.2159201. URL https://doi.org/10.1109/TSMCC.2011.2159201 [robot] [assistive] (Q1)

30. Juan G. Victores, Santiago Martinez, Alberto Jardón, and Carlos Balaguer. Robot-aided tunnel inspection and maintenance system by vision and proximity sensor integration. *Automation in Construction*, 20:629–636, 2011a. ISSN 09265805. doi: 10.1016/j.autcon.2010.12.005. URL https://doi.org/10.1016/j.autcon.2010.12.005 [robot] [construction] (Q1)

Patents (2)

1. Alberto Jardón Huete, Santiago Martínez, Juan G. Victores, Carlos Balaguer, Rafael Portero, and Marc Martí. Sistema y método para la verificación de la trayectoria de un tunel, 2014. URL https://invenes.oepm.es/InvenesWeb/detalle?referencia=P201330794 [robot] [construction]

2. Juan G. Victores, Santiago Martinez, Alberto Jardón, and Carlos Balaguer. Tool and method for the automatic remote application of strips of fibre-reinforced polymer tape, comprising the dispensing of epoxy adhesive, 2011b. URL https://www.google.im/patents/WO2011138481A1?cl=en [robot] [construction]

Book Chapters (9)

1. Juan G. Victores, Elisabeth Menendez, and Carlos Balaguer. Tunnel structural inspection and assessment using an autonomous robotic system. In *Infrastructure Robotics: Methodologies, Robotic Systems and Applications*, pages 185–203. Wiley Online Library, 2024. doi: 10.1002/9781394162871.ch9. URL https://doi.org/10.1002/9781394162871.ch9 [robot] [construction]

2. Raul Fernandez-Fernandez, Juan G. Victores, and Carlos Balaguer. New trends and challenges in the automatic generation of new tasks for humanoid robots. In *Robocity16: Open Conference on Future Trends in Robotics*, pages 169–176. CSIC, 5 2016. ISBN 978-84-608-8452-1. URL https://www.robocity2030.org/events/event/evento-esp-2-2/ [robot] [xgnitive: cgda]

3. David Estevez, Juan G. Victores, and Carlos Balaguer. A new generation of entertainment robots enhanced with augmented reality. In *RoboCity16: Open Conference on Future Trends in Robotics*, pages 129–136. CSIC, 5 2016a. URL https://www.robocity2030.org/events/event/evento-esp-2-2/ [robot] [video-game]

4. David Estevez, Juan G. Victores, and Carlos Balaguer. Future trends in perception and manipulation for unfolding and folding garments. In *RoboCity16: Open Conference on Future Trends in Robotics*, pages 333–340. CSIC, 5 2016b. URL https://www.robocity2030.org/events/event/evento-esp-2-2/ [robot] [textiles: folding]

5. Roberto Montero, Juan G. Victores, Elisabeth Menéndez, and Carlos Balaguer. The robot-spect eu project: Autonomous robotic tunnel inspection. In *Robocity2030 13th Workshop EU robotic projects results*, pages 91–100. 2015b. URL https://www.robocity2030.org/events/event/13th-robocity2030-workshop/ [robot] [construction]

6. Santiago Morante, Juan G. Victores, Santiago Martínez, and Carlos Balaguer. Force-sensorless friction and gravity compensation for robots. In *Advances in Intelligent Systems and Computing*, volume 418. Springer International Publishing, 2015c. ISBN 9783319271484. doi: 10.1007/978-3-319-27149-1_5. URL https://doi.org/10.1007/978-3-319-27149-1_5 [robot] [modelling]

7. Alberto Jardón, Félix R. Cañadillas, Juan G. Victores, Santiago Martínez, and Carlos Balaguer. A review of eight years of ceabot contest: A national wide mini humanoids competition. In Manuel A. Armada, Alberto Sanfeliu, and Manuel Ferre, editors, *ROBOT2013: First Iberian Robotics Conference*, pages 41–52. Springer International Publishing, 2014a. ISBN 978-3-319-03652-6. doi: 10.1007/978-3-319-03653-3_4. URL https://doi.org/10.1007/978-3-319-03653-3_4 [robot] [educational]

8. Juan G. Victores, Félix R. Cañadillas, Santiago Morante, Alberto Jardón, and Carlos Balaguer. Assistive robot multi-modal interaction with augmented 3d vision and dialogue. In Manuel A. Armada, Alberto Sanfeliu, and Manuel Ferre, editors, *ROBOT2013: First Iberian Robotics Conference*, pages 209–217. Springer International Publishing, 2014b. ISBN 978-3-319-03412-6. doi: 10.1007/978-3-319-03413-3_15. URL https://doi.org/10.1007/978-3-319-03413-3_15 [robot] [assistive]

9. Carlos Balaguer and Juan G. Victores. Robotic tunnel inspection and repair. In *Technology Innovation in Underground Construction*, pages 445–460. CRC Press, 2010. URL https://www.crcpress.com/product/isbn/9780415551052 [robot] [construction]

Conference Proceedings (83)

1. Franciscso José Naranjo Campos, Juan G. Victores, Almudena Alcaide, Jesica Rivero, and Carlos Balaguer. Temporal expert-based reward learning for inverse reinforcement learning. In *Jornadas de Automática*. Universidade da Coruna, 9 2026a. doi: 10.17979/ja-cea.2026.47.13710. URL https://doi.org/10.17979/ja-cea.2026.47.13710 [robot] [learning]

2. Álvaro Santos-García, Bartek Łukawski, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Yarp and ros 2 genicam middleware for visual servoing applications. *Jornadas de Automática*, 9 2026a. doi: 10.17979/ja-cea.2026.47.13836. URL https://doi.org/10.17979/ja-cea.2026.47.13836 [robot] [teleoperation]

3. Bartek Łukawski, Grigor Hovhannisyan Mkhitaryan, Lucía Menéndez Reques, Iñaki Ureta Sanz, Juan G. Victores, and Alberto Jardón Huete. Integrating yarp and ros 2 for web-based and embedded robot guis. *Jornadas de Automática*, 9 2026a. doi: 10.17979/ja-cea.2026.47.13858. URL https://doi.org/10.17979/ja-cea.2026.47.13858 [robot] [teleoperation]

4. Álvaro Santos-García, Bartek Łukawski, Dorin Copaci, Juan G. Victores, and Carlos Balaguer. Bilateral haptic teleoperation bridging yarp and ros 2 ecosystems. In *Simposios del Comité Español de Automática (CEA)*, 5 2026b. doi: 10.64117/simposioscea.v2i2.216. URL https://doi.org/10.64117/simposioscea.v2i2.216 [robot] [teleoperation]

5. Bartek Łukawski, Alba Olano Díaz, Alfredo González Villasante, Edwin D. Oña, Juan G. Victores, and Alberto Jardón. External force-torque sensor feedback on the abb gofa collaborative robot. In *Simposios del Comité Español de Automática (CEA)*, 5 2026b. doi: 10.64117/simposioscea.v2i2.215. URL https://doi.org/10.64117/simposioscea.v2i2.215 [robot] [assistive]

6. Francisco José Naranjo Campos, Juan Carlos González Victores, Carlos Balaguer, and Alberto Jardón. Single-demonstration imitation with residual reinforcement learning for dual-arm robotic bottle opening. In *Simposios del Comité Español de Automática (CEA)*, 5 2026b. doi: 10.64117/simposioscea.v2i2.209. URL https://doi.org/10.64117/simposioscea.v2i2.209 [robot] [assistive]

7. Bartek Łukawski, Edwin D. Oña, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Assessing student satisfaction and system usability in industrial robotics education. In *Simposios del Comité Español de Automática (CEA)*, 2 2026c. doi: 10.64117/simposioscea.v2i1.150. URL https://doi.org/10.64117/simposioscea.v2i1.150 [robot] [educational]

8. Francisco J. Naranjo-Campos, Juan G. Victores, Jesús Tornero, and Carlos Balaguer. A robotic assistant for supporting hospitalized patients through object delivery. In *IBERDISCAP*, 11 2025a. doi: 10.1007/978-3-032-18746-8. URL https://doi.org/10.1007/978-3-032-18746-8 [robot] [assistive]

9. Ignacio Montesino, Aroa Bachiller Gomez, Juan G. Victores, Carlos Balaguer, and Alberto Jardon. Quasi-god object and geodesically restricted 6-dof haptic forces for compliant constraints and low frequency simulation. In *International Conference on Intelligent Robots and Systems (IROS)*. IEEE/RSJ, 10 2025. doi: 10.1109/IROS60139.2025.11245986. URL https://doi.org/10.1109/IROS60139.2025.11245986 [robot] [assistive]

10. Bartek Łukawski, Miquel Martín, Carmen Menchén, Edwin D. Oña, Juan G. Victores, and Alberto Jardón. Espresso macchiato, por favore: collaborative robotic coffee-making for education. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 9 2025a. doi: 10.17979/ja-cea.2025.46.12274. URL https://doi.org/10.17979/ja-cea.2025.46.12274 [robot] [assistive]

11. Bartek Łukawski, Mercedes Rebollo, Ángel Gilabert, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Yarp cartesian controller layers over ros 2 for teleoperation and web apps. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 9 2025b. doi: 10.17979/ja-cea.2025.46.12252. URL https://doi.org/10.17979/ja-cea.2025.46.12252 [robot] [assistive]

12. Francisco J. Naranjo-Campos, Juan G. Victores, Ana Calzada-Garcia, and Carlos Balaguer. Manipulación robótica mediante aprendizaje por refuerzo inverso con características basadas en trayectorias expertas. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 9 2025b. doi: 10.17979/ja-cea.2025.46.12175. URL https://doi.org/10.17979/ja-cea.2025.46.12175 [robot] [assistive]

13. Bartek Łukawski, Edwin Daniel Oña, Alberto Jardón, Juan G. Victores, and Carlos Balaguer. Development of educational applications with abb gofa collaborative robot using externally guided motion. In *International Conference on Control, Automation and Diagnosis (ICCAD)*, 7 2025c. doi: 10.1109/ICCAD64771.2025.11099395. URL https://doi.org/10.1109/ICCAD64771.2025.11099395 [robot] [educational]

14. Bartek Łukawski, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Interaction with a humanoid robot through a conversational interface using deepseek. In *Simposio CEA de Robótica, Bioingeniería, Visión Artificial y Automática Marina (RBVM)*, 6 2025d. doi: 10.64117/simposioscea.v1i1.65. URL https://doi.org/10.64117/simposioscea.v1i1.65 [robot] [teleoperation]

15. Francisco J. Naranjo-Campos, Juan G. Victores, Carlos Balaguer, and Alberto Jardon. Algebraic machine learning for robotic garment unfolding. In *Simposio CEA de Robótica, Bioingeniería, Visión Artificial y Automática Marina (RBVM)*, 6 2025c. doi: 10.64117/simposioscea.v1i1.35. URL https://doi.org/10.64117/simposioscea.v1i1.35 [robot] [textiles: folding]

16. Bartek Łukawski, Ignacio Montesino, Edwin Daniel Oña, Juan G. Victores, Carlos Balaguer, and Alberto Jardon. Towards the development of telepresence applications with tiago and tiago++ using a virtual reality headset. In *2025 IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*, pages 192–197, 4 2025e. doi: 10.1109/ICARSC65809.2025.10970173. URL https://doi.org/10.1109/ICARSC65809.2025.10970173 [robot] [teleoperation]

17. Ignacio Montesino, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Cartesian impedance control generalized to one-parameter splines. In *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 4701–4707, 2024a. doi: 10.1109/IROS58592.2024.10801939. URL https://doi.org/10.1109/IROS58592.2024.10801939 [robot] [assistive]

18. Alicia Herrera García-Mascaraque, Ignacio Montesino, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Integrator for musculoskeletal simulation in python. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 7 2024. URL https://doi.org/10.17979/ja-cea.2024.45.10802 [robot] [assistive]

19. Ignacio Montesino, Hugo Alonso Camara, Juan G. Victores, Carlos Balaguer, and Alberto Jardón. Geodesic restricted aruco-based positioning for vr rehabilitation robotics. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 7 2024b. URL https://doi.org/10.17979/ja-cea.2024.45.10964 [robot] [assistive]

20. Jaime Mas, Juan G. Victores, and Carlos Balaguer. Optimización de caminata con aprendizaje por refuerzo en humanoide teo. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 7 2024. URL https://doi.org/10.17979/ja-cea.2024.45.10950 [robot] [humanoid]

21. Francisco J. Naranjo-Campos, Ainhoa De Matías-Martínez, Juan G. Victores, José A. Gutiérrez Dueñas, Almudena Alcaide, and Carlos Balaguer. Robot tiago para servicio en cafetería. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 7 2024d. URL https://doi.org/10.17979/ja-cea.2024.45.10795 [robot] [assistive]

22. Bartek Łukawski, Alberto Rodríguez-Sanz, Elisabeth Menendez, Juan G. Victores, and Carlos Balaguer. A user-friendly point cloud processing pipeline for interfacing pcl with yarp. In *Jornadas de Automática*. Universidade da Coruña. Servizo de publicacións, 7 2024a. URL https://doi.org/10.17979/ja-cea.2024.45.10925 [robot] [teleoperation]

23. Ana Calzada, Bartek Łukawski, Juan G. Victores, and Carlos Balaguer. Teleoperation of the robot tiago with a 3d mouse controller. In *Actas del Simposio de Robótica, Bioingeniería y Visión por Computador: Badajoz, 29 a 31 de mayo de 2024*, pages 133–138. Universidad de Extremadura. Servicio de Publicaciones, 5 2024. ISBN 978-84-9127-262-5. URL https://hdl.handle.net/10662/21260 [robot] [teleoperation]

24. Bartek Łukawski, Alberto Rodríguez Sanz, Juan G. Victores, and Carlos Balaguer. An open-source implementation of a force-torque sensor data acquisition device for the humanoid robot teo. In *Actas del Simposio de Robótica, Bioingeniería y Visión por Computador: Badajoz, 29 a 31 de mayo de 2024*, pages 79–84. Universidad de Extremadura. Servicio de Publicaciones, 5 2024b. ISBN 978-84-9127-262-5. URL https://hdl.handle.net/10662/21260 [robot]

25. Johnny J. Yepez-Figueroa, Juan G. Victores, Alberto Jardón, and Carlos Balaguer. Diseño mecatrónico y construcción de un robot móvil omni-direccional de tres ruedas para transporte de carga en ambientes industriales. In *Actas del Simposio de Robótica, Bioingeniería y Visión por Computador: Badajoz, 29 a 31 de mayo de 2024*, pages 43–48. Universidad de Extremadura. Servicio de Publicaciones, 5 2024. ISBN 978-84-9127-262-5. URL https://hdl.handle.net/10662/21260 [robot] [design]

26. Francisco José Naranjo-Campos, Ainhoa de Matías Martínez, Juan G. Victores, José Antonio Gutiérrez Dueñas, Almudena Alcaide, and Carlos Balaguer. Detección y manipulación de botellas con el robot móvil manipulador tiago. In *Actas del Simposio de Robótica, Bioingeniería y Visión por Computador: Badajoz, 29 a 31 de mayo de 2024*, pages 37–42. Universidad de Extremadura. Servicio de Publicaciones, 5 2024e. ISBN 978-84-9127-262-5. URL https://hdl.handle.net/10662/21260 [robot] [assistive]

27. Bartek Łukawski, Juan G. Victores, and Carlos Balaguer. A generic controller for teleoperation on robotic manipulators using low-cost devices. In *XLIV Jornadas de Automática: libro de actas: Universidad de Zaragoza, Escuela de Ingeniería y Arquitectura, 6, 7 y 8 de septiembre de 2023, Zaragoza*, pages 785–788. Servizo de Publicacións. Universidade da Coruña, 9 2023. doi: 10.17979/spudc.9788497498609.785. URL https://doi.org/10.17979/spudc.9788497498609.785 [robot] [assistive]

28. Sofia Hernández Pérez, Ignacio Montesino Valle, Juan G. Victores, Edwin Daniel Oña, and Alberto Jardón Huete. Ros2 gesture classification pipeline towards gamified neuro-rehabilitation therapy. In *XLIV Jornadas de Automática: libro de actas: Universidad de Zaragoza, Escuela de Ingeniería y Arquitectura, 6, 7 y 8 de septiembre de 2023, Zaragoza*, pages 611–616. Servizo de Publicacións. Universidade da Coruña, 9 2023. doi: 10.17979/spudc.9788497498609.611. URL https://doi.org/10.17979/spudc.9788497498609.611 [robot] [assistive]

29. Ainhoa De Matías-Martínez, Francisco J. Naranjo-Campos, Juan G. Victores, and Carlos Balaguer. Planificador global se(2) para la navegación de robots móviles manipuladores en ros. In *Jornadas de Robótica y Bioingeniería*, pages 85–90. CEA UPM CSIC, 6 2023. ISBN 978-84-09-51892-0. doi: 10.20868/UPM.book.74896. URL https://doi.org/10.20868/UPM.book.74896 [robot] [planning]

30. Bartek Łukawski, Ignacio Montesino Valle, Juan G. Victores, Alberto Jardón, and Carlos Balaguer. An inverse kinematics problem solver based on screw theory for manipulator arms. In *XLIII Jornadas de Automática: libro de actas: 7, 8 y 9 de septiembre de 2022, Logroño (La Rioja)*, pages 864–869. Servizo de Publicacións da UDC, 9 2022. doi: 10.17979/spudc.9788497498418.0864. URL https://doi.org/10.17979/spudc.9788497498418.0864 [robot] [kinematics]

31. Ignacio Montesino Valle, Bartek Łukawski, Juan G. Victores, Alberto Jardón Huete, and Carlos Balaguer. Entorno de gym basado en impedancia para el robot colaborativo iiwa de cara a interacción humano robot. In *XLIII Jornadas de Automática: libro de actas: 7, 8 y 9 de septiembre de 2022, Logroño (La Rioja)*, pages 762–769. Servizo de Publicacións da UDC, 9 2022. doi: 10.17979/spudc.9788497498418.0762. URL https://doi.org/10.17979/spudc.9788497498418.0762 [robot] [assistive]

32. Francisco José Naranjo-Campos, Ainhoa de Matías Martínez, Juan G. Victores, Nicolás Álvarez López, Almudena Alcaide Raya, and Carlos Balaguer. Manipulación de objetos dirigida a la asistencia de personas con movilidad reducida. In *XLIII Jornadas de Automática: libro de actas: 7, 8 y 9 de septiembre de 2022, Logroño (La Rioja)*, pages 798–803. Servizo de Publicacións da UDC, 9 2022. doi: 10.17979/spudc.9788497498418.0798. URL https://doi.org/10.17979/spudc.9788497498418.0798 [robot] [assistive]

33. Raul Fernandez-Fernandez, Marco Aggravi, Paolo Robuffo Giordano, Juan G. Victores, and Claudio Pacchierotti. Neural style transfer with twin-delayed ddpg for shared control of robotic manipulators. In *2022 International Conference on Robotics and Automation (ICRA)*, pages 4073–4079, 5 2022b. doi: 10.1109/ICRA46639.2022.9812245. URL https://doi.org/10.1109/ICRA46639.2022.9812245 [robot] [xgnitive: cgda]

34. Jennifer J. Gago, Bartek Łukawski, Juan G. Victores, and Carlos Balaguer. Under-actuation modelling in robotic hands via neural networks for sign language representation with end-user validation. In Paulo, Camacho David, Yin Hujun Analide Cesar, and Novais, editors, *Intelligent Data Engineering and Automated Learning – IDEAL 2020*, pages 239–251. Springer International Publishing, 2020. ISBN 978-3-030-62365-4. URL https://doi.org/10.1007/978-3-030-62365-4_23 [robot] [sign-language]

35. J. Enrique Sierra-Garcia, Matilde Santos, and Juan G. Victores. Neural controller of uavs with inertia variations. In *IDEAL 2019*, pages 169–177. Springer International Publishing, 11 2019. ISBN 978-3-030-33617-2. doi: 10.1007/978-3-030-33617-2_19. URL https://doi.org/10.1007/978-3-030-33617-2_19 [robot] [control]

36. Jennifer J. Gago, Valentina Vasco, Bartek Łukawski, Ugo Pattacini, Vadim Tikhanoff, Juan G. Victores, and Carlos Balaguer. Sequence-to-sequence natural language to humanoid robot sign language. In *EUROSIM 2019*, page 44. ARGESIM, 7 2019b. ISBN 978-3-901608-92-6. doi: 10.11128/arep.58. URL https://www.doi.org/10.11128/arep.58 [robot] [sign-language]

37. David Estevez, Juan G. Victores, Raul Fernandez-Fernandez, and Carlos Balaguer. Towards clothes hanging via cloth simulation and deep convolutional networks. In *EUROSIM 2019*, page 35. ARGESIM, 7 2019. ISBN 978-3-901608-92-6. doi: 10.11128/arep.58. URL https://www.doi.org/10.11128/arep.58 [robot] [textiles: hanging]

38. Raul Fernandez-Fernandez, Juan G. Victores, David Estevez, and Carlos Balaguer. Quick, stat!: A statistical analysis of the quick, draw! dataset. In *EUROSIM 2019*, page 27. ARGESIM, 7 2019. ISBN 978-3-901608-92-6. doi: 10.11128/arep.58. URL https://www.doi.org/10.11128/arep.58 [robot] [xgnitive: drl]

39. Jennifer J. Gago, Bartek Łukawski, Juan G. Victores, and Carlos Balaguer. A study on the effects of an embodied humanoid robot representing sign language. In *Gesture-Sign Workshop*, pages 26–28, 2019c. URL https://calc.ff.cuni.cz/en/gswp19_programme/ [robot] [sign-language]

40. Raul Fernandez-Fernandez, Juan G. Victores, David Estevez, and Carlos Balaguer. Robot imitation through vision, kinesthetic and force features with online adaptation to changing environments. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 6546–6551. IEEE, 2018b. ISBN 978-1-5386-8094-0. doi: IROS.2018.8593724. URL https://doi.org/10.1109/IROS.2018.8593724 [robot] [xgnitive: cgda]

41. David Estevez, Juan G. Victores, Raul Fernandez-Fernandez, and Carlos Balaguer. Robotic ironing with 3d perception and force/torque feedback in household environments. In IEEE, editor, *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, volume 2017-Septe, pages 6484–6489. IEEE, 2017a. ISBN 9781538626818. doi: 10.1109/IROS.2017.8206556. URL https://doi.org/10.1109/IROS.2017.8206556 [robot] [textiles: ironing]

42. Elisabeth Menendez, Juan G. Victores, and Carlos Balaguer. Sistema robótico para la inspección y análisis estructural de túneles. In *Jornadas Nacionales de Robótica*, 2017a. URL https://jnr2017.ai2.upv.es/wp-content/uploads/2016/11/Programa-JNR2017.pdf [robot] [construction]

43. David Estevez, Juan G. Victores, and Carlos Balaguer. Horus: Inspección robotizada de los trajes de protección del personal sanitario de pacientes en aislamiento de alto nivel, incluido el Ébola. In *Jornadas Nacionales de Robótica*, 2017b [robot] [horus]

44. Elisabeth Menendez, Juan G. Victores, Roberto Montero, and Carlos Balaguer. Autonomous robotic system with tunnel inspection tool positioning. In *Proceedings of the International Symposium on Automation and Robotics in Construction (ISARC)*, pages 655–662, 2017b. URL https://doi.org/10.22260/ISARC2017/0091 [robot] [construction]

45. Roberto Montero, Elisabeth Menendez, Juan G. Victores, and Carlos Balaguer. Intelligent robotic system for autonomous crack detection and caracterization in concrete tunnels. In *2017 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2017*, pages 316–321. IEEE, 2017. ISBN 9781509062331. doi: 10.1109/ICARSC.2017.7964094. URL https://doi.org/10.1109/ICARSC.2017.7964094 [robot] [construction]

46. David Estevez, Raul Fernandez-Fernandez, Juan G. Victores, and Carlos Balaguer. Robotic ironing with a humanoid robot using human tools. In *2017 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2017*, pages 134–139. IEEE, 2017c. ISBN 9781509062331. doi: 10.1109/ICARSC.2017.7964065. URL https://doi.org/10.1109/ICARSC.2017.7964065 [robot] [textiles: ironing]

47. David Estevez, Raul Fernandez-Fernandez, Juan G. Victores, and Carlos Balaguer. Improving and evaluating robotic garment unfolding: A garment-agnostic approach. In *2017 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2017*. IEEE, 2017d. ISBN 9781509062331. doi: 10.1109/ICARSC.2017.7964077. URL https://doi.org/10.1109/ICARSC.2017.7964077 [robot] [textiles: folding]

48. Raul Fernandez-Fernandez, David Estevez, Juan G. Victores, and Carlos Balaguer. Improving cgda execution through genetic algorithms incorporating spatial and velocity constraints. In *2017 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2017*, pages 290–295. IEEE, 2017a. ISBN 9781509062331. doi: 10.1109/ICARSC.2017.7964090. URL https://doi.org/10.1109/ICARSC.2017.7964090 [robot] [xgnitive: cgda]

49. Raul Fernandez-Fernandez, David Estevez, Juan G. Victores, and Carlos Balaguer. Reducing the number of evaluations required for cgda execution through particle swarm optimization methods. In *2017 IEEE International Conference on Autonomous Robot Systems and Competitions, ICARSC 2017*, pages 284–289. IEEE, 2017b. ISBN 9781509062331. doi: 10.1109/ICARSC.2017.7964089. URL https://doi.org/10.1109/ICARSC.2017.7964089 [robot] [xgnitive: cgda]

50. David Estevez, Juan G. Victores, and Carlos Balaguer. A lightweight finite state machine c++ library aimed at seamless integration with robotic middlewares. In *Towards Humanoid Robots OS Workshop at HUMANOIDS 2016*, 11 2016c. URL https://roboticslab-uc3m.github.io/workshop-humanoids2016 [robot] [video-game]

51. Konstantinos Loupos, Angelos Amditis, Anastasios Doulamis, Philippe Chrobocinski, Juan G. Victores, Max Wietek, Panagiotis Panetsos, Alberto Roncaglia, Stephanos Camarinopoulos, Vassileios Kallidromitis, Dimitrios Bairaktaris, Nikolaos Komodakis, and Rafa Lopez. Integrated robotic solution for tunnel structural evaluation and characterization – robo-spect ec project. In *International Conference on Smart Infrastructure and Construction (ICSIC)*, 2016. URL https://www.icsic.eng.cam.ac.uk/programme/programme21june/view [robot] [construction]

52. David Estevez, Juan G. Victores, Santiago Morante, and Carlos Balaguer. Towards robotic garment folding: A vision approach for fold detection. In *IEEE International Conference on Autonomous Robot Systems and Competitions (ICARSC)*, 5 2016d. doi: 10.1109/ICARSC.2016.65. URL https://doi.org/10.1109/ICARSC.2016.65 [robot] [textiles: folding]

53. Santiago Morante, Juan G. Victores, and Carlos Balaguer. Automatic demonstration and feature selection for robot learning. In *IEEE International Conference on Humanoid Robot - Humanoids*, pages 428–433. IEEE, 2015d. doi: 10.1109/HUMANOIDS.2015.7363569. URL https://doi.org/10.1109/HUMANOIDS.2015.7363569 [robot] [xgnitive: cgda]

54. Konstantinos Loupos, Angelos Amditis, Christos Stentoumis, Juan G. Victores, Philippe Chrobocinski, Alberto Roncaglia, Stephanos Camarinopoulos, Nikos Komodakis, and Rafael Lopez. Robotic system with intelligent vision for tunnel structural assessment - system architecture – the robospect ec project. In *Third conference on smart monitoring, assessment and rehabilitation of civil structures (SMAR)*, 2015. URL https://data.smar-conferences.org/SMAR_2015_Proceedings/html/L.html [robot] [construction]

55. David Estevez, Juan G. Victores, Santiago Morante, and Carlos Balaguer. Robot devastation: Using diy low-cost platforms for multiplayer interaction in an augmented reality game. In *7th International Conference on Intelligent Technologies for Interactive Entertainment (INTETAIN)*, volume 15, pages 32–36. IEEE, 6 2015b. ISBN 9781479983773. URL https://ieeexplore.ieee.org/document/7325482/ [robot] [video-game]

56. Konstantinos Loupos, Angelos Amditis, Christos Stentoumis, Philippe Chrobocinski, Juan G. Victores, Max Wietek, Panagiotis Panetsos, Alberto Roncaglia, Stephanos Camarinopoulos, Vassilis Kalidromitis, Dimitris Bairaktaris, Nikos Komodakis, and Rafa Lopez. Robotic intelligent vision and control for tunnel inspection and evaluation - the robinspect ec project. In *IEEE International Symposium on Robotic and Sensors Environments (ROSE)*, 2014. URL https://doi.org/10.1109/ROSE.2014.6952986 [robot] [construction]

57. Alberto Jardón, Santiago Martínez, Juan G. Victores, and Carlos Balaguer. Extended range guidance system for the teleoperation of microtunnelling machines. In *31st International Symposium on Automation and Robotics in Construction and Mining (ISARC 2014)*, pages 837–846. University of Technology, Sydney, 2014b. URL https://www.isarc2014.org/pdfs/ISARC2014_Proceedings-NewISBN.pdf [robot] [construction]

58. Carlos Balaguer, Roberto Montero, Juan G. Victores, Santiago Martínez, and Alberto Jardón. Towards fully automated tunnel inspection: A survey and future trends. In Quang Ha, Xuesong Shen, and Ali Akbarnezhad, editors, *Proceedings of the 31st International Symposium on Automation and Robotics in Construction and Mining (ISARC 2014)*, pages 19–33. University of Technology, Sydney, 2014. URL https://www.isarc2014.org/pdfs/ISARC2014_Proceedings-NewISBN.pdf [robot] [construction]

59. Santiago Morante, Juan G. Victores, Alberto Jardón, and Carlos Balaguer. On using guided motor primitives to execute continuous goal-directed actions. In *Proceedings - IEEE International Workshop on Robot and Human Interactive Communication*, volume 2014-Octob. IEEE, 2014a. doi: 10.1109/ROMAN.2014.6926320. URL https://doi.org/10.1109/ROMAN.2014.6926320 [robot] [xgnitive: cgda]

60. Santiago Morante, Juan G. Victores, Alberto Jardón, and Carlos Balaguer. Action effect generalization, recognition and execution through continuous goal-directed actions. In *ICRA IEEE International Conference on Robotics and Automation*. IEEE, 2014b. doi: 10.1109/ICRA.2014.6907098. URL https://doi.org/10.1109/ICRA.2014.6907098 [robot] [xgnitive: cgda]

61. Juan G. Victores, Santiago Morante, Alberto Jardón, and Carlos Balaguer. Creación de tareas de asistencia robótica mediante la interacción multimodal. In *Congreso Iberoamericano de Tecnologias de*

Apoyo a la Discapacidad Iberdiscap, pages 21–24, 2013a. URL https://iberdiscap2013.pucmm.edu.do/programa [robot] [assistive]

62. Juan G. Victores, Santiago Morante, Alberto Jardón, and Carlos Balaguer. Towards robot imagination through object feature inference. In *IROS IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 5694–5699. IEEE, 2013b. doi: 10.1109/IROS.2013.6697181. URL https://doi.org/10.1109/IROS.2013.6697181 [robot] [xgnitive: ris]

63. Martin F. Stoelen, Virginia F. Tejada, Juan G. Victores, Alberto Jardón, Fabio Bonsignorio, and Carlos Balaguer. Adaptive collision-limitation behavior for an assistive manipulator. In *Proceedings of IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2013)*, pages 1143–1148, 2013. doi: 10.1109/IROS.2013.6696494. URL https://doi.org/10.1109/IROS.2013.6696494 [robot] [control]

64. Félix R. Cañadillas, Alberto Jardón, Carlos Balaguer, and Juan G. Victores. Diseño preliminar de interfaces de realidad aumentada para el robot asistencial asibot. In *V Congreso Internacional de Diseño, Redes de Investigación y Tecnología para todos (DRT4ALL)*, pages 213–219, 2013. URL https://www.discapnet.es/Castellano/areastematicas/tecnologia/DRT4ALL/EN/DRT4ALL1013/Paginas/Introduction.aspx [robot] [assistive]

65. Juan G. Victores, Santiago Morante, Alberto Jardón, and Carlos Balaguer. "give me the red can": Assistive robot task creation through multi-modal interaction. In *V Congreso Internacional de Diseño, Redes de Investigación y Tecnología para todos (DRT4ALL)*, 2013c. URL https://www.discapnet.es/Castellano/areastematicas/tecnologia/DRT4ALL/EN/DRT4ALL1013/Paginas/Introduction.aspx [robot] [assistive]

66. Miguel González-Fierro, Miguel Angel Maldonado, Juan G. Victores, Santiago Morante, and Carlos Balaguer. Object tagging for human-robot interaction by recolorization using gaussian mixture models. In *Robocity2030 12th Workshop Robotica Cognitiva*, 2013. URL https://www.researchgate.net/publication/269631653_Object_Tagging_for_Human-Robot_Interaction_by_Recolorization_Using_Gaussian_Mixture_Models [robot] [vision]

67. Juan G. Victores, Santiago Morante, Alberto Jardón, and Carlos Balaguer. Semantic action parameter inference through machine learning methods. In *Robocity2030 12th Workshop Robotica Cognitiva*, pages 35–46, 2013d. URL https://www.researchgate.net/publication/269631454_Semantic_Action_Parameter_Inference_through_Machine_Learning_Methods [robot][xgnitive: ris]

68. Juan G. Victores, Santiago Morante, Miguel González-Fierro, and Carlos Balaguer. Augmented reality and social interaction platform through multirobot design. In *Robocity2030 11th Workshop Robots Sociales*, pages 131–143, 2013e. URL https://www.researchgate.net/publication/269630253_Augmented_reality_and_social_interaction_platform_through_multirobot_design [robot] [assistive]

69. Alberto Jardón, Santiago Martínez, Juan G. Victores, M. Marti, and Carlos Balaguer. Extended range guidance system for micro-tunnelling machine. In *International Symposium for Automation and Robotics in Construction 2012 (ISARC/Gerontechnology 2012)*, 2012b. URL https://gerontechnology.info/index.php/journal/article/view/1814 [robot] [construction]

70. Alberto Jardón, Juan G. Victores, Martin F Stoelen, Santiago Martínez, and Carlos Balaguer. Experimental evaluation of assistive robots in virtual domestic scenarios. In *International Symposium for Automation and Robotics in Construction (ISARC/Gerontechnology 2012)*, 2012c. URL https://www.gerontechnology.info/index.php/journal/article/view/gt.2012.11.02.489.00 [robot] [construction]

71. Juan Gonzalez-Gomez, Juan G. Victores, Alberto Valero-Gomez, and Mohammed Abderrahim. Motion control of differential wheeled robots with joint limit constraints. In *IEEE International Conference on Robotics and Biomimetics (ROBIO)*, pages 596–601, 2011. doi: 10.1109/ROBIO.2011.6181351. URL https://doi.org/10.1109/ROBIO.2011.6181351 [robot] [control]

72. Carlos Balaguer, Alberto Jardón, Concepción Monje, Fabio Bonsignorio, Martin F Stoelen, Santiago Martínez, and Juan G. Victores. Sultan: Simultaneous user learning and task execution, and its application in assistive robotics. In Kazuyoshi Wada, Machiel Van der Loos, and Loredana Zollo, editors, *Workshop on New and Emerging Technologies in Assistive Robotics at IROS 2011*, pages 6–8, 2011. URL https://www.iros2011.org/WorkshopsAndTutorialsProceedings/MW3/iros11ws_assistive_robotics.pdf [robot] [assistive]

73. Alberto Jardón, Martin F Stoelen, Virginia Fernández, Juan G. Victores, Santiago Martínez, Carlos Balaguer, and Fabio Bonsignorio. Aplicación de teoría de la información para el modelado y cuantificación de la interacción persona-robot. In *IV Congreso Internacional de Diseño, Redes de Investigación y Tecnología para todos (DRT4ALL)*, pages 36–45. Fundación ONCE, 2011b. URL https://www.discapnet.es/Castellano/areastematicas/tecnologia/DRT4ALL/ES/DRT4ALL2011/Documents/Libro-de-actas-pdf [robot] [assistive]

74. Juan G. Victores, Alberto Jardón, Santiago Morante, Martin F Stoelen, Santiago Martinez, and Carlos Balaguer. Interacción humano-robot a través de interfaces en la nube. In *Robocity2030 9th Workshop, Robots colaborativos e interaccion humano-robot*, pages 75–92, 2011c. ISBN 978-84-7484-238-8. URL https://www.researchgate.net/publication/269574247_Interaccion_humano-robot_a_traves_de_interfaces_en_la_nube [robot] [assistive]

75. Martin F Stoelen, Alberto Jardón, Virginia Fernández, Juan G. Victores, Santiago Martinez, Fabio Bonsignorio, and Carlos Balaguer. Methodologies for experimental evaluation of assistive robotics hri. In *Robocity2030 9th Workshop. Robots colaborativos e interacción humano-robot*, 2011. ISBN 978-84-614-5558. URL https://www.researchgate.net/publication/269574315_Methodologies_for_Experimental_Evaluation_of_Assistive_Robotics_HRI [robot] [assistive]

76. Juan G. Victores, Santiago Martinez, Alberto Jardón, and Carlos Balaguer. R.a.t.i.m. sistema robótico de inspección y mantenimiento de túneles. In *Robocity2030 8th Workshop. Robots de Exteriores.*, pages 373–382, 2010a. ISBN 978-84-614-5558. URL https://www.researchgate.net/publication/269574005_RATIM_Sistema_Robtico_de_Inspeccin_y_Mantenimiento_de_Tneles [robot] [construction]

77. Miguel González-Fierro, Alberto Jardón, Santiago Martinez, Martin F Stoelen, Juan G. Victores, and Carlos Balaguer. Educational initiatives related with the ceabot contest. In *Proceedings of Int. Conf. on SIMULATION, MODELING and PROGRAMMING for AUTONOMOUS ROBOTS (SIMPAR2010) Workshops*, pages 649–658, 2010. URL https://asrob.uc3m.es/svn/ASROBrepo/Documentos/articulos/29-TeachingRobotics.pdf [robot] [educational]

78. Juan G. Victores, Alberto Jardón, Martin F Stoelen, Santiago Martinez, and Carlos Balaguer. Asibot assistive robot with vision in a domestic environment. In *Robocity2030 7th Workshop Visión en Robótica*, 2010b. ISBN 84-693-6777-3. URL https://www.researchgate.net/publication/269573982_ASIBOT_assistive_robot_with_vision_in_a_domestic_environment [robot] [vision]

79. Martin F Stoelen, Alberto Jardón, Juan G. Victores, Carlos Balaguer, and Fabio Bonsignorio. Information metrics for assistive human-in-the-loop cognitive systems. In *Workshop on Good Experimental Methodology in Robotics and Replicable Robotics Research, Robotics Science and Systems (RSS)*, pages 1–6. Zaragoza. Spain., 2010a. URL https://www.heronrobots.com/EuronGEMSig/Downloads/Zaragoza/rss2010_gemrrrr.pdf [robot] [control]

80. Martin F Stoelen, Alberto Jardón, Fabio Bonsignorio, Juan G. Victores, Concepción Monje, and Carlos Balaguer. Towards an enabling multimodal interface for an assistive robot. In *Workshop on Mutimodal Human-Robot Interfaces, IEEE International Conference on Robotics and Automation (ICRA)*, 2010b. URL https://www.researchgate.net/publication/269573950_Towards_an_Enabling_Multimodal_Interface_for_an_Assistive_Robot [robot] [assistive]

81. Juan G. Victores, Alberto Jardón, Fabio Bonsignorio, Martin F Stoelen, and Carlos Balaguer. Benchmarking usability of assistive robotic systems: Methodology and application. In *Workshop on the Role of Experiments in Robotic Research, IEEE International Conference on Robotics and Automation (ICRA)*, 2010c. URL https://www.heronrobots.com/EuronGEMSig/downloads/Anchorage/ICRA10-experiments_final_uc3m.pdf [robot] [assistive]

82. Juan G. Victores, Santiago Martinez, Alberto Jardón, and Carlos Balaguer. Robot-aided tunnel inspection and maintenance system. In *26th International Symposium on Automation and Robotics in Construction (ISARC)*, pages 420–426, 2009. URL https://www.iaarc.org/publications/fulltext/Robot-aided_Tunnel_Inspection_and_Maintenance_System.pdf [robot] [construction]

83. Alberto Jardón, Juan G. Victores, Martin F Stoelen, Santiago Martinez, and Carlos Balaguer. Asibot assistive robot in a domestic environment. In *Proceedings of the 2nd International Conference on PErvsive Technologies Related to Assistive Environments (PETRA)*, pages 611–614. ACM Press, 2009. ISBN 9781605584096. doi: 10.1145/1579114.1579175. URL https://doi.org/10.1145/1579114.1579175 [robot] [assistive]

Conference Videos (2)

1. Juan G. Victores, Santiago Morante, Alberto Jardón, and Carlos Balaguer. On using humanoid robot imagination to perform the shortened token test. In *IEEE RAS International Conference on Humanoid Robots (Humanoids 2014)*, page 172. IEEE, 2014c. doi: 10.1109/HUMANOIDS.2014.7041355. URL https://youtu.be/KFHFW9dJfzA [robot] [xgnitive: ris]

2. Santiago Morante, Juan G. Victores, Santiago Martinez, and Carlos Balaguer. Sensorless friction and gravity compensation. In *IEEE RAS International Conference on Humanoid Robots (Humanoids 2014)*, page 265. IEEE, 2014c. doi: 10.1109/HUMANOIDS.2014.7041370. URL https://www.dailymotion.com/video/x2vjrfs [robot] [modelling]

Late Breaking Results Poster (1)

1. Juan Miguel Garcia-Haro, Santiago Martinez, Edwin Daniel Oña Simbaña, and Juan G. Victores. Whole-body postural control approach based on multiple zmp evaluation in humanoid robots. In *International Conference on Intelligent Robots and Systems (IROS)*. IEEE/RSJ, 11 2019 [robot] [humanoid]

PhD Thesis (Author)

1. Juan G. Victores. *Robot Imagination System*. PhD thesis, Universidad Carlos III de Madrid, 2014. URL https://hdl.handle.net/10016/19834

PhD Thesis (Advisor)

1. Raul Fernandez-Fernandez. *Action Generalization in Humanoid Robots Through Artificial Intelligence*

With Learning From Demonstration. PhD thesis, Universidad Carlos III de Madrid, 9 2021. URL https://hdl.handle.net/10016/33536

2. David Estevez. *Robotic System for Garment Perception and Manipulation*. PhD thesis, Universidad Carlos III de Madrid, 11 2020. URL https://hdl.handle.net/10016/32186

3. Santiago Morante. *Continuous Goal-Directed Actions: Advances in Robot Learning*. PhD thesis, Universidad Carlos III de Madrid, 3 2016. URL https://hdl.handle.net/10016/23459

Research Stays (2)

Oct. 2018 – Jan. 2019	**The University of Manchester**. Cognitive Robotics Lab. Estancia de 3 meses destinado a la investigación del uso de Deep Learning para el estudio de conceptos abstractos con el robot iCub (Manchester, UK).
Sept. 2011 – Dic. 2011	**Istituto Italiano di Tecnologia**. Department of Robotics, Brain and Cognitive Sciences. Estancia de 3 meses destinado a la investigación del uso de Support Vector Machines y Gaussianas mixtas para el control en fuerza del robot iCub (Génova, Italia).

Workshop Organizer (2)

1. Juan G. Victores, Lorenzo Natale, Eiichi Yoshida. Towards Humanoid Robots OS. HUMANOIDS. Cancun, Mexico. Nov 15. 2016. https://roboticslab-uc3m.github.io/workshop-humanoids2016/

2. Angelos Amditis, Konstantinos Loupos, Juan G. Victores. Autonomous Robotic Systems for Inspection and Structural Assessment of Civil Underground Infrastructures. European Robotics Forum (ERF). Ljubljana, Slovenia. Mar 22. 2016. https://www.eu-robotics.net/robotics_forum/upload/digest_1-96_without_emails_250ppi1.pdf

Talks (3)

1. Juan G. Victores, Bartek Łukawski, Jennifer J. Gago. Proyecto TEO. TTD 2019 DEMOS DE TECNOLOGÍA ACCESIBLE. https://youtu.be/JJ9QhE2rzbY?t=2403

2. Juan G. Victores. XGNITIVE: Avances hacia la generalización avanzada de acciones y sistemas de imaginación en robótica. Technology Festival (Techfest). Universidad Rey Juan Carlos (URJC). 2017. https://www.eventbrite.es/e/registro-technology-festival-urjc-2017-28838850779?aff=es2#

3. Angelos Amditis, Juan G. Victores, Fedi Francesco. Welcome and Introduction. Autonomous Robotic Systems for Inspection and Structural Assessment of Civil Underground Infrastructures. European Robotics Forum (ERF). Ljubljana, Slovenia. Mar 22. 2016. https://www.eu-robotics.net/robotics_forum/upload/digest_1-96_without_emails_250ppi1.pdf